

Bywaf Usage Guide

Overview

Bywaf is a highly-auditable Python 3 commandlet framework for authorized web application and network testing workflows. It presents a Metasploit-like interactive shell, loads commandlets from plugins, and connects commandlets through a SQLite-backed event bus.

The core idea is simple: one commandlet discovers something and publishes it as an event; another commandlet consumes that event and publishes the next result. For example, **hostscanner** can publish live hosts, **portscanner** can consume those hosts and publish open ports, and HTTP commandlets can consume open ports and probe web services.

Use Bywaf only on systems and networks where you have explicit authorization.

Evolving framework design notes are tracked in DESIGN.md.

Installation

During development, run Bywaf from the repository root:

```
cd bywaf
python3 -m bywaf --help
python3 -m bywaf repl
```

For an editable local install:

```
python3 -m pip install -e .
bywaf --help
bywaf repl
```

The project metadata defines a console script named **bywaf**, so packaged installations can expose Bywaf as a normal command instead of requiring `python3 -m bywaf`.

Bywaf can also be embedded as a Python library. A local GUI or web service should use the public session facade instead of scraping REPL output:

```
from pathlib import Path
from bywaf import BywafSession

session = BywafSession.open(Path(".bywaf/bywaf.sqlite3"))
session.run("hostscanner 127.0.0.1")
hosts = session.events(topic="host.found")
jobs = session.jobs()
```

The host and port scanner commandlets use **nmap** through a Python adapter. A local **nmap** binary is required for real scans. The adapter prefers **nmaplib**, then

python-nmap, then nmapthon, then libnmap.

Dependency summary:

nmap	required for hostscanner and portscanner
nmaplib/python-nmap/etc.	Python nmap adapter; Bywaf tries supported adapters
sqlcipher3-binary	optional Python SQLCipher driver for encrypted DBs
sqlcipher	optional system SQLCipher tooling/library
scapy	optional helper library for future packet plugins

Bywaf plugins are intended to wrap useful external tools and normalize their results into the central event database. That removes manual handoffs such as copying hosts to notes or intermediate files: one plugin can discover hosts, another can consume those host events, and later plugins can continue the workflow from the same stored data.

Execution-time plugin variables are scoped by commandlet. A plugin uses `context.vars.get("name")` for its own variables and cannot enumerate another plugin's variables through that API. Explicit global variables use `context.vars.get_global("name")`. When a commandlet run starts, Bywaf snapshots the effective commandlet and global variables into SQLite under that `command_run_id`; `show run=<id>` displays the captured variables so runs remain auditable and reproducible even when session variables change later.

Plugins that need interpreter-owned actions use request events instead of direct method calls. For example, a plugin can publish `shell.prompt.requested`; the foreground REPL validates the request and records either `shell.prompt.updated` or `framework.request.denied` for auditability. Plugins also declare intended capabilities on `CommandSpec`; Bywaf records audit-only `plugin.capability.used` and `plugin.capability.missing` events so operators can compare intended behavior with actual behavior. Plugin event-bus access should go through `context.events`, which audits `db.read:<topic>` and `db.write:<topic>` capability usage. Raw `context.db` access is retained for privileged/internal framework commandlets during the transition and audits `db.raw`.

Encrypted databases require SQLCipher support. On Debian or Ubuntu, install the SQLCipher library and use the optional Python extra:

```
sudo apt install sqlcipher libsqlcipher-dev
python3 -m pip install -e '.[sqlcipher]'
```

Starting Bywaf

Start the interactive shell:

```
bywaf
```

or, from a source checkout:

```
python3 -m bywaf
```

Create or open the default database with SQLCipher encryption:

```
bywaf --encrypt  
bywaf --database client.sqlite3 --encrypt
```

Run one command non-interactively:

```
bywaf run ls  
bywaf run cat README.md  
bywaf run 'hostscanner 127.0.0.1 | portscanner'
```

Simple `run` commands do not need quotes. Use quotes when the command contains shell metacharacters such as `|`, `&`, `>`, or spaces that must be preserved inside a single argument.

REPL Basics

The REPL prompt is:

```
bywaf>
```

Commandlets can be run directly:

```
bywaf> ls  
bywaf> cat README.md  
bywaf> hostscanner 127.0.0.1
```

Built-in commands manage the shell and the event database:

```
help  
help <command>  
plugins  
cmds  
vars  
history  
job <list|show|cancel|kill>  
pipeline <list|show|cancel|kill>  
cancel <job=id|pipeline=id>  
kill [--force] <job=id|pipeline=id>  
jobs  
runs  
topics  
db <status|path|checkpoint|vacuum|new|encrypt|decrypt|rekey>  
show <topic|job=id|run=id|pipeline=id>  
load <resource>  
save <resource>  
exit
```

`help <command>` shows the same help as `<command> --help` for commandlets.

Commandlets

A commandlet is an executable unit exposed by a plugin. Each commandlet declares its name, description, arguments, consumed topics, and emitted topics.

Examples:

```
bywaf> help hostscanner
bywaf> hostscanner --help
bywaf> portscanner --help
```

Many scanning commandlets support `-s` or `--silent` to suppress console alerts. Commandlets request alerts through the framework using the database; the framework validates the request, stores a structured `console.alert` event, and prints it unless silent mode is active:

```
hostscanner <hostscanner-...>: discovered host 127.0.0.1
portscanner <portscanner-...>: discovered port 127.0.0.1:80/tcp
```

Commandlets can also declare tab-completion behavior for their arguments and options. For example, `ls [path]`, `cat <path>`, and `less <path>` get filename completion because those commandlets declare path/file completion in their plugin specs. Other completion specs include `topic`, `run`, `pipeline`, `job`, and `plugin`, so plugin authors can make hand-typed commands much easier to complete correctly.

Plugin authors should use `context.output()`, `context.table()`, `context.alert()`, `context.progress()`, `context.page_file()`, and `context.process` instead of direct `print()` calls, direct terminal control, or direct subprocess calls. These helpers keep terminal output, progress, and external tool execution auditable and make the same commandlets usable from a future GUI or web frontend.

Progress is separate from findings. A finding is durable evidence such as `host.found` or `port.open`; progress is operational state such as “42% through the TCP scan.” Commandlets report progress through structured events:

```
context.progress_started(phase="tcp_scan", total=1000, unit="ports")
context.progress(phase="tcp_scan", current=420, total=1000, unit="ports")
context.progress_completed(phase="tcp_scan", current=1000, total=1000, unit="ports")
```

Bywaf enforces progress throttling in the framework. Configure it with global session variables:

```
bywaf> vars global.progress.min-interval-ms=250
bywaf> vars global.progress.min-percent-delta=1
```

Audit logs are stored as SQLite events. Use `audit show ...` to inspect them and `audit export file=audit.jsonl`, `audit export file=audit.pdf`, or `audit`

export file=audit.sqlite3 to hand off a copy.

```
bywaf> audit show topic=console.alert since=20260517 until=20260518
bywaf> audit export file=audit.pdf since=run=<command-run-id>
bywaf> audit export --encrypt file=audit.sqlite3
bywaf> audit export --encrypt file=audit.pdf
```

Unqualified `since=` and `until=` audit bounds default to `time:.` Encrypted SQLite audit exports use SQLCipher. Encrypted PDF export uses `pikepdf` when available, otherwise the external `qpdf` command.

Plugins

A plugin provider groups related commandlets. The `plugins` command lists loaded providers:

```
bywaf> plugins
discovery
http
network
os
runtime
storage
```

The `cmds` command lists commandlets grouped by provider:

```
bywaf> cmds
discovery
  hostscanner
http
  http_headers
  http_probe
network
  portscanner
os
  cat
  less
  ls
runtime
  job
storage
  db
```

Bundled plugins are listed in `bywaf/plugins/plugins.json`. Adding a plugin file is not enough to load it by default; add its dotted path to that config.

Load an additional plugin by name from `.bywaf/plugins`:

```
bywaf> load plugin=myplugin
```

Load a plugin from an explicit filesystem path:

```
bywaf> load plugin=./plugins/myplugin
bywaf> load plugin=~/.bywaf-plugins/myplugin
```

Pipelines

Pipelines connect commandlets with `|`. Prefix the expression with `name:` to name the pipeline without consuming commandlet arguments:

```
bywaf> hostscanner 127.0.0.1 | portscanner
bywaf> client subnet scan: hostscanner 127.0.0.1 | portscanner
```

The runner executes each stage in order. Events emitted by one stage are passed to the next stage as input. Events are also stored in SQLite with a pipeline ID and command run ID.

This model allows downstream commandlets to consume only the output relevant to the current pipeline, rather than every historical event in the database.

Attach a new background commandlet to an existing pipeline:

```
bywaf> pipeline attach <pipeline-id> portscanner run=<producer-run-id> since=beginning
bywaf> pipeline attach <pipeline-id> http_probe since=now
```

The attach selectors are orthogonal:

- `<pipeline-id>` chooses the pipeline the new command run joins.
- `run=<producer-run-id>` optionally narrows input to one upstream producer run.
- `since=beginning` replays matching historical events, then listens for new events.
- `since=now` ignores historical events and starts from the current event high-water mark.

If `run=` is omitted, the attached commandlet reads matching events from the whole pipeline.

Runtime Names

Name the current command run with a stage-local `name=` selector:

```
bywaf> hostscanner 127.0.0.1 name=localhost sweep
```

Name or inspect runtime entities after they exist:

```
bywaf> name run=<command-run-id> localhost sweep
bywaf> name pipeline=<pipeline-id> client subnet scan
bywaf> name job=<job-id> background listener
bywaf> name run=<command-run-id>
```

The explicit keyed form is `text=`, for example `name run=<id> text=localhost sweep`.

Assigned names appear in `runs`, `pipelines`, and `jobs` listings.

Framework Notes

Any commandlet stage can include a framework-level `note=` selector. The runner strips it before the commandlet receives arguments and records an audited `note.attached` event with the job, pipeline, and command-run IDs.

```
bywaf> hostscanner 10.0.0.0/24 note=client-approved internal subnet
```

If `note=` is the last selector in a stage, it consumes the rest of that stage without requiring quotes:

```
bywaf> hostscanner targets note=scope approved | portscanner note=top ports
```

Review attached notes with the `note` commandlet. Output and file exports use timestamp-first lines:

```
bywaf> note run=<command-run-id>
bywaf> note pipeline=<pipeline-id>
bywaf> note job=<job-id> file=notes.txt
bywaf> note add run=<command-run-id> text=follow-up note
```

Notes are append-only. Adding another note creates another timestamped `note.attached` event instead of replacing earlier notes.

Encrypted Artifacts

Artifacts are evidence files attached to a run, pipeline, or job. Artifact bodies are stored in a separate encrypted SQLCipher database next to the main database, using the main encrypted database passphrase for the session. The main database stores timestamped provenance events such as `artifact.attached` and `artifact.exported`; it does not store artifact bodies.

Start Bywaf with an encrypted database before attaching artifacts:

```
bywaf --encrypt
```

Attach one or more files:

```
bywaf> artifact attach run=<command-run-id> file=snapshot.html file=headers.txt
bywaf> artifact attach pipeline=<pipeline-id> file=report.json note=initial report
```

List, save, and verify artifacts:

```
bywaf> artifact list run=<command-run-id>
bywaf> artifact replace artifact=1 file=snapshot-v2.html
bywaf> artifact remove artifact=1
```

```
bywaf> artifact save artifact=1 file=snapshot.html
bywaf> artifact save run=<command-run-id> dir=artifacts/
bywaf> artifact verify pipeline=<pipeline-id>
```

Use `file=` when saving exactly one artifact. Use `dir=` when saving a set. If `file=` matches multiple artifacts, Bywaf reports that clearly and asks you to use `dir=` instead.

At-File Arguments

Any commandlet argument can use framework-level at-file expansion. Expansion happens before the commandlet receives its argument list and is audited as a framework argument expansion. When the active database is encrypted and artifact storage is available, the expanded input file is also attached as an encrypted provenance artifact.

```
bywaf> hostscanner @lines:targets.txt
bywaf> hostscanner 192.168.1.1-255 except=@lines:do-not-scan.txt
bywaf> http_probe @target.txt
bywaf> echo @@literal-at-sign
```

Supported forms:

- `@file:` read the file as one text argument
- `@raw:file:` read the file as one text argument
- `@lines:file:` expand each non-empty line into a separate argument
- `@@value:` pass `@value` literally

Scanner commandlets support `except=` for exclusion lists. Values may be comma separated or file-backed through `@lines:.`

Tab completion preserves these prefixes while completing filesystem paths.

Variable Expansion

Bywaf expands `$variables` before commandlets receive their argument list. Unquoted and double-quoted variables expand; single-quoted variables are passed literally.

```
bywaf> use hostscanner
bywaf> vars targets=192.168.1.1 192.168.1.2
bywaf> hostscanner $targets
bywaf> hostscanner "$targets"
bywaf> hostscanner '$targets'
```

Resolution checks the exact variable name first, then the active commandlet scope, then `global..` For example, `$targets` in `hostscanner` checks `targets`,

`hostscanner.targets`, and `global.targets`. Variable expansion is audited as `framework.variable.expanded`.

Command Continuation And Sequences

Use a trailing backslash to continue a command across physical lines in the interactive shell or in scripts:

```
bywaf> hostscanner \  
... 192.168.0.0/24
```

Separate multiple commands with semicolons when they should run sequentially:

```
bywaf> vars target=127.0.0.1; vars; topics
```

Semicolons inside quotes are preserved as part of the argument text.

Background Execution

Append `&` to background a commandlet or pipeline:

```
bywaf> hostscanner 192.168.0.1-255 &
```

Normal commandlet execution is job-audited through the database. Foreground commandlets run in-process but still record `job.requested`, `job.claimed`, `job.started`, and `job.finished` or `job.failed`. Background jobs use the same job lifecycle, but a worker process claims and runs the queued job. Foreground management commands such as `db ...` and `job ...` run directly.

Stage-level backgrounding works inside pipelines:

```
bywaf> hostscanner 192.168.0.1-255 & | portscanner &
```

In that example, `portscanner` listens for `host.found` rows created by the immediately upstream `hostscanner` run in the same pipeline. It does not consume unrelated `host.found` rows from older scans.

List active jobs, or all jobs with an explicit active marker:

```
bywaf> job list  
bywaf> job list --all  
bywaf> jobs --all
```

Show one job:

```
bywaf> job show <id>
```

Soft-cancel a job so commandlets that check cancellation can exit cleanly:

```
bywaf> job cancel <id>
```

Hard-stop a job process:

```
bywaf> job kill <id>
bywaf> job kill --force <id>
```

Pipelines can be inspected and controlled the same way:

```
bywaf> pipeline list
bywaf> pipeline list --all
bywaf> pipeline show <id>
bywaf> pipeline cancel <id>
bywaf> pipeline kill <id>
```

`job list`, `runs`, and `pipeline list` show active runtime state by default. Use `--all` to include historical entries. When jobs are listed with `--all`, Bywaf prefixes history rows with `[active]`, `[in progress]`, `[failed]`, or `[completed]` so stale and finished jobs, runs, and pipelines are visible without looking active. Set `vars global.listing.active-format=long` to print an indented marker such as `[active since 2026-05-17 04:30:00 UTC]` beneath the entity row; set it to `short` for the compact prefix form.

For hand-typed control, `cancel` and `kill` also accept selector syntax:

```
bywaf> cancel job=<id>
bywaf> cancel pipeline=<id>
bywaf> kill job=<id>
bywaf> kill --force pipeline=<id>
bywaf> pause job=<id>
bywaf> pause run=<command-run-id>
bywaf> resume --listonly pipeline=<id>
bywaf> resume --listonly run=<command-run-id>
bywaf> stop --hard job=<id>
```

`jobs` remains as a convenience alias for `job list`, and `pipelines` remains as a convenience alias for `pipeline list`.

Database and Event Model

Bywaf stores events in SQLite. The default database is:

```
.bywaf/bywaf.sqlite3
```

SQLite is used in WAL mode. Inserts are committed immediately; there is no separate document-style save step. On shutdown, Bywaf checkpoints the WAL using `PRAGMA wal_checkpoint(TRUNCATE)` so WAL contents are folded back into the main database file.

List known event topics:

```
bywaf> topics
```

Show recent events for a topic:

```
bywaf> show host.found
```

```
bywaf> show port.open
```

List command runs:

```
bywaf> runs
```

Show events by command run or pipeline:

```
bywaf> show run=<command-run-id>
```

```
bywaf> show pipeline=<pipeline-id>
```

Save a database snapshot:

```
bywaf> save db=snapshot.sqlite3
```

Save an encrypted snapshot:

```
bywaf> save --encrypt db=snapshot.sqlite3
```

Inspect or maintain the active database:

```
bywaf> db status
```

```
bywaf> db path
```

```
bywaf> db checkpoint
```

```
bywaf> db vacuum
```

Create a fresh database and switch the active session to it:

```
bywaf> db new
```

```
bywaf> db new --file=client.sqlite3
```

```
bywaf> db new --encrypt --file=client.sqlite3
```

```
bywaf> db new --force --file=client.sqlite3
```

Without `--file`, `db new` creates a timestamped database under `.bywaf/db/`. With `--file`, it refuses to overwrite an existing file. Add `--force` to move the existing database and SQLite sidecar files to timestamped `.bak-*` names before creating the new DB. `--encrypt` forces SQLCipher encryption and prompts twice for a passphrase.

The session variable `db.encryption=sqlcipher` makes `db new` encrypted by default:

```
bywaf> vars db.encryption=sqlcipher
```

```
bywaf> db new
```

Convert the active database in place:

```
bywaf> db encrypt
```

```
bywaf> db rekey
```

```
bywaf> db decrypt
```

`db encrypt` converts the active plaintext database to SQLCipher and prompts twice for a new passphrase. `db rekey` changes the passphrase for an encrypted

database. `db decrypt` exports the active encrypted database back to plaintext SQLite after an explicit **YES** confirmation.

Switch to another database:

```
bywaf> load db=snapshot.sqlite3
```

If the database is encrypted, Bywaf prompts for its passphrase when loading it. Passphrases are kept in process memory only and are not written to config or history files.

Resource Files

Bywaf keeps default state in:

`.bywaf/`

Important files:

```
.bywaf/bywaf.sqlite3
.bywaf/config.json
.bywaf/history.bywaf
.bywaf/plugins/
```

Resource resolution rules are consistent:

```
plugin=<name>    -> .bywaf/plugins/<name>
script=<name>    -> ./<name>
db=<name>        -> ./<name>
config=<name>    -> ./<name>
history=<name>   -> ./<name>
```

Explicit paths are used as filesystem paths for every resource type:

```
./name
../name
~/name
/absolute/name
```

Examples:

```
bywaf> load script=scan.bywaf
bywaf> load script=./scripts/scan.bywaf
bywaf> save config=session.json
bywaf> load config=session.json
bywaf> save history=session-history.bywaf
bywaf> load history=session-history.bywaf
```

Variables

List variables:

```
bywaf> vars
```

Set a variable:

```
bywaf> vars name=value
```

Common examples:

```
bywaf> vars http_probe.cookie-file=/tmp/cookies.txt
bywaf> vars history.timestamp-format=%Y-%m-%d %H:%M:%S %Z
bywaf> vars hostscanner.targets=192.168.1.1-255
bywaf> hostscanner
```

Use a commandlet context to make short variable assignments target that commandlet:

```
bywaf> use hostscanner
bywaf> vars targets=192.168.1.1-255
bywaf> use global
```

For commandlets that opt into variable defaults, explicit command-line arguments take precedence over commandlet variables, and commandlet variables take precedence over built-in defaults. For example, `hostscanner 127.0.0.1` ignores `hostscanner.targets`, while `hostscanner` falls back to it.

Save variables:

```
bywaf> save config=config.json
```

Load variables:

```
bywaf> load config=config.json
```

Config files are JSON objects containing session variables.

History

Every non-empty REPL command is appended to:

```
.bywaf/history.bywaf
```

History lines are stored as commands followed by a timestamp comment:

```
hostscanner 127.0.0.1 # 2026-05-12 10:15:30 EDT
```

The `history` command shows only commands from the current REPL invocation, with timestamps displayed first for easier scanning:

```
bywaf> history
2026-05-17 10:00:00 EDT  plugins
```

Limit session history by timestamp with `since=` and `until=`. Unqualified values default to `time:` and use `yyyymmdd[HH[MM[SS]]]`:

```
bywaf> history since=20260517 until=20260518
bywaf> history since=time:202605171000 until=time:202605171059
```

The persistent history file can be viewed with the OS commandlets:

```
bywaf> cat .bywaf/history.bywaf
bywaf> less .bywaf/history.bywaf
```

The persistent history file stays script-friendly: timestamps are stored after commands as comments, so history lines can be copied into a script file.

Change the timestamp format:

```
bywaf> vars history.timestamp-format=%Y/%m/%d %H:%M:%S %Z
```

Scripts

Scripts are text files with one command expression per line:

```
# scan local host
hostscanner 127.0.0.1
portscanner --from-topic host.found
```

Blank lines and lines beginning with `#` are ignored. Inline comments are also allowed after whitespace:

```
plugins # list loaded plugin providers
```

Run a script:

```
bywaf> load script=scan.bywaf
```

Bundled Commandlets

os

`ls` lists local files:

```
bywaf> ls
bywaf> ls bywaf/plugins
```

`cat` prints a local text file:

```
bywaf> cat README.md
```

`less` opens the system `less` pager when running interactively:

```
bywaf> less README.md
```

Use `/` to search inside `less`, arrow keys or paging keys to scroll, and `q` to quit.

discovery

`hostscanner` discovers live hosts with `nmap`:

```
bywaf> hostscanner 127.0.0.1
bywaf> hostscanner 192.168.0.1-255
bywaf> hostscanner 192.168.1-3.1-255
```

It emits `host.found` events.

network

`portscanner` scans hosts for open ports:

```
bywaf> portscanner 127.0.0.1
bywaf> portscanner --ports 22,80,443 127.0.0.1
```

If `--ports` is omitted, `nmap` uses its normal default top-port behavior. It emits `port.open` events.

Use listen mode to consume newly inserted hosts:

```
bywaf> portscanner --listen
```

http

`http_headers` performs HTTP HEAD requests and emits `http.headers` events:

```
bywaf> http_headers example.com
bywaf> http_headers --ssl true example.com
```

`http_probe` probes HTTP endpoints and emits `http.endpoint` events:

```
bywaf> http_probe https://example.com/
```

For authorized session-aware testing, it can use cookies:

```
bywaf> vars http_probe.cookie-file=/path/to/cookies.txt
bywaf> http_probe https://example.com/
bywaf> http_probe --firefox-profile ~/.mozilla/firefox/<profile>
```

Common Workflows

Scan one host for live status:

```
bywaf> hostscanner 127.0.0.1
```

Scan one host for ports:

```
bywaf> portscanner 127.0.0.1
```

Discover hosts and scan their ports:

```
bywaf> hostscanner 192.168.0.1-255 | portscanner
```

Run discovery and port scanning in the background:

```
bywaf> hostscanner 192.168.0.1-255 & | portscanner &
```

Probe HTTP services after port scanning:

```
bywaf> hostscanner 127.0.0.1 | portscanner --ports 80,443 | http_probe
```

Save the current database:

```
bywaf> save db=scan-results.sqlite3
bywaf> save --encrypt db=scan-results.sqlite3
```

Save variables and session history:

```
bywaf> save config=session.json
bywaf> save history=session.bywaf
```

Troubleshooting

If `python` is not available, use `python3`:

```
python3 -m bywaf
```

If scanning fails with an `nmap` error, verify that `nmap` is installed and that the selected Python `nmap` binding is available.

If socket creation is denied, the process may be running inside a sandbox or without the privileges needed by the selected scan type.

If a command is unknown, check loaded commandlets:

```
bywaf> cmds
```

If a plugin does not load, verify its directory structure and that it defines a `plugin()` factory returning a commandlet.

SQLite WAL files such as `.sqlite3-wal` and `.sqlite3-shm` are normal. Bywaf checkpoints the WAL on shutdown.

Developer Notes

The main package layout is:

<code>bywaf/app.py</code>	REPL and built-in commands
<code>bywaf/runner.py</code>	parsing, pipelines, foreground/background execution
<code>bywaf/db.py</code>	SQLite event store
<code>bywaf/plugin.py</code>	commandlet protocol and specs
<code>bywaf/registry.py</code>	plugin discovery and loading
<code>bywaf/completion.py</code>	readline completion
<code>bywaf/plugins/</code>	bundled plugin providers
<code>tests/</code>	unit tests

Run tests:

```
python3 -m unittest discover -s tests
```

Add a commandlet by defining a class with a `CommandSpec` and a `run()` method, then expose it through a `plugin()` factory. Add bundled commandlets to `bywaf/plugins/plugins.json` when they should load by default.

Commandlets can declare completion metadata with `ArgumentSpec`, `OptionSpec(..., completion=CompletionSpec(...))`, or an optional custom `complete(context, args, prefix)` method. See `PLUGIN_AUTHOR_GUIDE.md` for a walkthrough and a small working example.

Reference

Useful built-ins:

```
help [command]
plugins
cmds
vars [name=value]
history
job <list|show|cancel|kill>
pipeline <list|show|cancel|kill|attach>
cancel <job=id|pipeline=id>
kill [--force] <job=id|pipeline=id>
name <run=id|pipeline=id|job=id> [name text]
note [add] <run=id|pipeline=id|job=id> [text=note|file=path]
artifact <attach|list|remove|replace|save|verify> [artifact=id|run=id|pipeline=id|job=id] [1
jobs
pipelines
runs
topics
show <topic>
show job=<id>
show run=<id>
show pipeline=<id>
db <status|path|checkpoint|vacuum|new|encrypt|decrypt|rekey>
load plugin=<resource>
load script=<resource>
load db=<resource>
load config=<resource>
load history=<resource>
save db=<resource>
save --encrypt db=<resource>
save config=<resource>
save history=<resource>
```

```
prompt [pattern]  
exit
```

Common event topics:

```
host.found  
port.open  
http.headers  
http.endpoint  
job.requested  
job.claimed  
job.started  
job.finished  
job.failed
```